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An Integrated AI Framework for Industrial Risk Management: Predicting Raw Material Price Regimes and Carbon Compliance Risk Using Concept Drift Detection

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Certificate

This is to certify that the Dissertation titled **"An Integrated AI Framework for Industrial Risk Management: Predicting Raw Material Price Regimes and Carbon Compliance Risk Using Concept Drift Detection"** is a bona fide record of the work carried out **by Chandan D,** Reg. No. **22SSDS415012** in partial fulfilment of the requirements for the award of a Bachelor's Degree of M. S. Ramaiah University of Applied Sciences in the Department of Data Sciences and Analytics.

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Declaration

"An Integrated AI Framework for Industrial Risk Management: Predicting Raw Material Price Regimes and Carbon Compliance Risk Using Concept Drift Detection"

The dissertation is submitted in partial fulfilment of academic requirements for the Bachelors' Degree of M. S. Ramaiah University of Applied Sciences in the Department of Data Sciences and Analytics. This dissertation is a result of my own investigation. All sections of the text and results, which have been obtained from other sources, are fully referenced. I understand that cheating and plagiarism constitute a breach of university regulations, hence this dissertation has been passed through plagiarism check.

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Chapter – 1

Introduction

INDUSTRY CONTEXT: MOTHERSON GROUP AND CIM TOOLS PRIVATE LIMITED

The Samvardhana Motherson Group: From Silver Trading to Global Manufacturing

Few corporate origin stories in Indian industry are as improbable, or as instructive, as that of the Samvardhana Motherson Group. In 1975, Vivek Chaand Sehgal and his mother, Shrimati Swaran Lata Sehgal, founded the company in Delhi with an initial investment of approximately fifteen US dollars. The name Motherson itself, suggested by Sehgal's father, was chosen to represent the bond of trust between a mother and her son — a value system that, as the company grew into a multinational conglomerate, remained at the centre of its corporate identity.

The move to the silver trade was only the start, taking a hard left turn when a competitor's business went under, jeopardising the operation. Seeking to find another way forward, Sehgal spotted a much more sustainable business model: manufacturing components for electrical wiring in India, which were the raw materials of the country's emerging automotive industry. As soon as the first car of domestic market "Maruti 800" was introduced in the year 1983, there was a sudden and great demand for auto harness wiring. Sehgal realized that the harness making was underway then in the country and visited Japan for the same, but due to the samples showing lack of quality, he had to invest and produce to Japanese standards for this business. It was educative – take the setback, work with a better technical combination, and repeat till perfection – Motherson culture and practice was born, which will hold for the next decades.

In year 1986 the group's publicly listed flagship Motherson Sumi Systems Limited (MSSL) was established along with Sumitomo Wiring Systems Ltd. of Japan. For Motherson, the decision to list in the Indian stock exchanges was part of his strategy; Sehgal recognised that the listing process would also drive professionalisation of the way things are managed and capital flowing in on better terms would give him greater competitiveness vis-à-vis other companies, bidding for contracts that demanded financial transparency. They started to publish 5-year strategic plans since 1995 functioning as a structured ambition for the otherwise opportunistically growing business.

The Acquisition Strategy: Building a Global Tier-1 Supplier

As Motherson transformed to become a global automotive Tier-1 supplier from being an Indian automotive supplier, they had a unique approach of doing an acquisition. What was universally true of all the different Indian conglomerates of the era was the search for acquisitions as a way for diversification, and Motherson's was disciplined to it - get a company with existing customer relationships, technical capabilities or manufacturing footprints that would otherwise be tough, if not impossible to get up and running organically within the ambit of the next five year plan.

The example is set by the first international acquisition in Ireland on 2002 – Wexford Electronics. When an existing supplier couldn't meet obligations, this European customer had a supply chain crisis. During a difficult period for the Company, Motherson was able to intervene by purchasing the manufacture plant and transporting five hundred tonnes of machinery from Ireland to Sharjah within 10-weeks whilst keeping the customers' supply lines from Texas opened. The programme revealed some of the attributes that would emerge as traits of the Motherson to come: think on one's feet attitude, the attitude of being the partner you are expected to be.

Significant strides towards the international league of automotive suppliers were made by the two acquisitions in 2009 (Motherson acquired Visiocorp and renamed it Samvardhana Motherson Reflectec or SMR and also in 2011 when they acquired SMP which is a leading polymer automotive module manufacturing company). SMR positioned Motherson as the leader of the rear-view mirror systems in the world and build a relationship with the world's top car manufacturers. SMP, a company about the same size as the entire

Motherson Group at the time of acquisition, virtually doubled the size of the company the very next day when it became one of the biggest suppliers of dashboards, door panels and bumper systems for premium auto Original Equipment Manufacturers such as BMW, Mercedes-Benz and Volkswagen. When Motherson first dreamed big in the late 1990s, by articulating its 5-year plans as a true global automotive supplier, he was perhaps a little tongue-in-cheek, but by the early '10s that had certainly become reality.

There were three guidelines always followed in making acquisitions by the group. Firstly, acquired firms' management, where possible, and independence of operation were retained, creating what amounted to a federated group structure which preserved the entrepreneurial spirit of the sub-unit group firms. Second, acquisitions were evaluated by their 'fit' with Motherson's existing customers and their ability to chart the capability matrix--not from a financial engineering point of view. Third, they undertook every acquisition in good, conservative fashion with the acquisition's leverage not so high when they were forced to put their money on what they eventually acquired. Motherson got the investors the credibility of open communication (in the annual reports and other investor products) that a pure financial acquirer never scores with the industrial partners.

Diversification Beyond Automotive: Vision 2025 and the Aerospace Pivot

At that time Motherson Group was a revenue size matched by no one else in India with its work force spread in 41 countries in the hundreds of thousands. The biggest business for the group has been the automotive business – and in 2020, Sehgal proposes an ambitious plan to change the company's leverage and footprint – to enter into four new businesses under Vision 2025 – Aerospace, Health and Medical, Logistics Solutions and Technology and Industrial Solutions! Along with growth, as was the thinking, an Motherson-sized automotive supplier was going to be highly at risk from some of the industry's structural shifts: electric vehicles and a plan to drastically cut the parts required for regular vehicle engines. Similarly, a diversified revenue base (across industrial sectors) will mitigate cyclical risks in addition to providing a new capability building area.

The most important and the best-staged diversification move was the aerospace decision. Looking on the outside, the difference between aerospace and automotive supply isn't so obvious as it does get for quality, regulatory environment, customer relationship dynamics, aerospace lead times and capital intensity of aerospace supply all differ considerably from automotive. Rather than attempt to enter the aerospace supply chain from scratch, Motherson looked for an established player in the Indian aerospace manufacturing ecosystem with existing OEM relationships, proven quality management systems, and a management team willing to operate in partnership with a large conglomerate. CIM Tools Private Limited was that company.

CIM Tools Private Limited: Building India’s Aerospace Precision Manufacturing Capability

CIM Tools Private Limited was founded in 1997 in Bengaluru, the city that has over time established itself as the undisputed aerospace manufacturing hub of India. Located initially in the Peenya Industrial Area before expanding to the KIADB Industrial Area SEZ Aerospace Park at Devanahalli, CIM Tools grew over more than two decades into one of the most technically credentialed aerospace component manufacturers in the Indian private sector.

The company’s core business is the supply of precision machined parts and sub-assemblies to commercial aerospace original equipment manufacturers. CIM Tools holds Tier 1 supplier status with Boeing — meaning it supplies directly to the aircraft manufacturer rather than through an intermediate supplier — and Tier 2 status with Airbus, covering all commercial variants of both manufacturers’ aircraft families. In practical terms, this means that CIM Tools components have flown on virtually every major commercial aircraft in service globally, from narrow-body single-aisle aircraft to wide-body long-haul jets.

The manufacturing capabilities developed by CIM Tools over its first two decades cover the full range of precision aerospace component production. Machined Parts Production: Aluminium alloys, titanium and aerospace quality steel are CNC'd to micron tolerances in order to manufacture high accuracy parts. The company has expertise in the process of forming and fabrication of structural parts in sheet metal. Through its subsidiary Aero Treatment Private Limited (ATPL), it delivers the focused surface treatment and finishing processes that are required by aerospace specifications: anodising, chromate conversion coating, paint systems and non-destructive testing.

Nickel superalloys and Titanium alloys are used in engine components and in a variety of structural parts and are known as extremely difficult-to-machine materials as part of the 2019 version of the JV with Lauak International. Most important, CIM Tools could also have its activities geared toward kit (or even equipping and sub-assembly), so it would no longer be selling individual parts to its partners but could move up the value chain to a 'Tier 1' position that Boeing provided it.

CIM Tools was not only a first order acquisition target due to its manufacturing capability, but due to its customer relationships and trajectory of its order book. During the October acquisition announcement of Motherson, CIM has a cumulative booking business potential of over INR 15 billion (more than USD 200 million) that it currently recognises with its OEM customers. The order book expanded by an additional 26 percent to around USD 252 million between October 2021 and completion of the acquisition in April 2022, a rate that was a testament to the strength of CIM Tools' market standing as well as the level of trust that CIM Tools' OEM partners placed in the operational continuity under new owners.

It was this vision, scaling up from individual parts manufactured by machining operations to larger assemblies, sub-assemblies and structures and building management systems that can successfully serve the exacting governance requirements of the biggest aerospace OEMs that saw the beginnings of CIM Tools by the three founders Mr Srikanth GS, Mr Umesh AS and Mr Vishwanath Deshpande. By 2021, that vision was underway, but the capital investments in the subsequent level of growth, a push for expanded facilities and the technology needed to be able to manufacture a more diverse component collection, highlighted a more resource-rich partner with global expertise and manufacturing capabilities.

The Acquisition: Structure, Rationale, and Post-Merger Integration

In October 2021, the majority stake of 55% was acquired by Motherson Sumi Systems Limited, the remaining stake of 45% being held by the three Founders who will continue to play operational roles with CIM Tools Private Limited. It was an intentional decision by Motherson when the founding shareholders decided to stay in the business after the acquisition, leading to the customer relationships, technical know-how and organisational culture that made CIM Tools valuable continuing to be alive and well on and off the acquisition. As a philosophy and throughout its acquisitions, Motherson has always worked on ‘acquisition of people first' and the acquisition of CIM Tools was done in the same spirit.

With this acquisition, Motherson made their 27th acquisition and the first acquisition in the aerospace industry. In Motherson's eyes, it was their four aerospace manufacturing centres, their proven Quality Management System (QMS), having successfully gone through the demanding auditors examination of Boeing, Airbus and even their supply chain, and their existing order book which offered immediate visibility into revenue in the new segment. For CIM Tools, the Motherson partnership provided access to Motherson's global manufacturing facilities, close ties with customers from around the world in the automotive and other industrial sectors, financial resources to expand facilities and operational know-how from decades of precision manufacturing in the auto industry and beyond.

This is the reason why the deal was officially confirmed in April 2022. But when it was finished, it was recognized that they had been able to enter an aerospace market at the "down side" of the business cycle. CIM Group has consolidated overall income to be about INR 548 million quarterly ending Q4, FY2022, which is in excess of the pre-COVID revenues of INR 2031 million (annual) with the publication in FY2020. The COVID-19 outbreak had caught global commercial aviation off-guard to a certain extent, having recently grounded fleets, and significantly impacted OEM production volumes. The path towards recovery, which the industry was on, was visible from the fact that the order book of CIM Tools was a realisable revenue, rather than contingent revenue, since the revenue started to recover in Q4 of FY2022.

The Chairman of Motherson, Mr Perugini, clearly enunciated the rationale of the deal. The closure of the acquisition will mark the lapse of the foundation of the Vision 2025 plan oriented around aerospace portfolio, Vivek Chaand Sehgal said, adding that such partnerships – spanning around the world and combining the scale of Motherson's and its technical prowess and contacts from an established sector leader – would form the backbone of the group's future diversification beyond automotive. In this case, the synergy logic was rather straightforward, as two organisations with complementary manufacturing capabilities, coupled with Motherson's global capabilities would enable the expanded entity to provide a greater range of solutions to customers within the aerospace industry.

Industrial Data Challenges: The Context for This Dissertation

My personal involvement in Motherson Aerospace/CIM Tools had been as an intern where I was involved in data analytics and industrial emission monitoring which are directly relevant to the research problem focused on in this dissertation for cascading to an operational level.

In fact, this aerospace manufacturing environment could be more complex from a first-cut perspective than the data may seem to describe, when it comes from a manufacturing facility like CIM Tools. Process monitoring provides a flow of sensor data from CNC machinery centres on the shopfloor; of spindle loads, vibration signatures, tool wear indicator and dimensional measurement of the produced parts. The number of inspection results reported in quality management systems, the number of non-conformance reports reported and rework rates. Different energy consumption for production at different equipment & areas are monitored and reflected impacting on the BEE PAT scheme reporting requirement for CIM Tools as a larger Industrial unit.

Large volumes of data is created and we're using very rudimentary methods to analyze that data and act. Data such as energy use was being gathered with diligence at the facility level – and in a monthly summary report – but not in real-time. A growing number of production planning decisions that have a high impact on the emission intensity of the production process (e.g. for which processes was it necessary to use machines, to use high energy surface treatment processes in front of or behind the part of the production process with the highest production peaks, whether to operate on power from the grid or on back-up power) were being made without any forward-looking indication of their compliance impact. The data existed. What was missing was the models for teaching compliance signals.

Similarly, the procurement function was exposed and the cost exposure was hard-wired to the commodity price markets relevant to the cost of the aerospace raw materials, including the cost of aluminium, titanium and some specialty alloys, without any systematic process in place to understand the commodity price market transition risk. The overall atmosphere of a shift in regimes would be based on information about the flow of information and additional contact with suppliers, and not so much on a quantitative early warning system. Proactive hedging and/or supply chain adjustment opportunities often were missed by the time proven regime changes were seen in procurement cost data.

This is because, unlike any academic article, the following observations in the shop-floor and the procurement office had a more direct impact on the design of this dissertation's research. The approach presented here in predicting the commodity price regimes and concept drift detection with carbon compliance risk monitoring is not an academically masochistic endeavor. It’s a push to create using available tools and data, what a facility like CIM Tools would really like to have: a ‘Uber for Risk Intelligence'.

The Broader Indian Aerospace Manufacturing Ecosystem

CIM tools are indeed a part of the ecosystem operating in the Indian aerospace manufacturing space where dynamic transformation is taking place – albeit currently at a fairly early developmental stage – when compared to the years-long established aerospace manufacturing countries of Europe and North America. Bengaluru has about 65 per cent of the aerospace industry activities in India and the city is home to Hindustan Aeronautics Limited (HAL), the Indian Space Research Organisation (ISRO), Airbus India, Boeing India Engineering, Collins Aerospace and other international and domestic players.

CIM Tools have invested in their technical prowess to manufacture components for the aerospace industry to international OEM standards, in addition to being compliant for defence applications, however, this has become a policy headwind from the Indian government's specific domestic content requirements for aerospace and defence procurement as part of the ‘Make in India' initiative and Defence Production Policy. For example, the formation of KIADB Aerospace Park located at Devanahalli where CIM Tools has its major operations is itself an outcome of such policy thrusts – an industrial estate with world-class infrastructure, providing SEZ benefits and also being in the vicinity of Kempegowda

International Airport, thereby making it conducive for the growth of the aerospace manufacturing cluster.

Regarding the risk management framework developed in this dissertation, there is a specific view related to the whole ecosystem. Aerospace manufacturers in India are exposed to global commodity markets for raw materials, particularly aluminium and titanium, whose prices are influenced by geopolitical developments, global supply dynamics, and the production schedules of the world's largest aerospace OEMs — all of which are outside the control of any individual Indian supplier. They are simultaneously subject to domestic regulatory frameworks, including the PAT scheme, and increasingly to the international regulatory requirements that flow from their customer relationships, including quality audits, supply chain sustainability requirements, and, as CBAM approaches full implementation, carbon reporting obligations to European customers. Monitoring and managing these intersecting risk exposures is precisely the challenge this dissertation's framework is designed to address.

1.1 Background and Motivation

Growing up around my father's fabrication workshop in Jamshedpur, I watched first-hand how a sudden spike in steel or gas prices could turn a profitable month into a scramble for working capital. The plant managers and small suppliers around that steel city had no early warning system. They absorbed the shock only after it had already arrived. That lived experience is the seed of this dissertation, and it is one I have carried into the research.

At a broader industrial scale, the same vulnerability plays out with far larger consequences. The same threat is present on an industrial level, but with much more grievous consequences. No clear trend in energy, base metals and chemical raw material prices. Economists call these other types of behavior representative of an alternate "regime with regimes": long periods of relative quiet followed by bouts of the most extreme turmoil and movement, with which the whole statistical order of law is different. When the market becomes a crisis regime, and everything turns upside down, the whole drive line of the factory may be upside down if a factory has tailored its procurement to a low price and low volatility time. The conventional hedging tools can be used only a limited amount, and they cannot warn the operator of an imminent transition.

Coinciding with the price volatility is another, newer and quickly growing challenge: carbon compliance. India's BEE has been working in the background setting energy-intensity benchmarks for the more than decade for implementation on large industrial units under this scheme, which is dubbed Perform, Achieve and Trade scheme. Not to mention, all of the EUs Carbon Border Adjustment Mechanism starts coming into financial force from 2026. This is no far-flung regulatory issue to Indian makers exporting to Europe. It is considered to be a short-term BS Risk. However, much of compliance monitoring in industry is more about 'tick the box at the end of the year' as opposed to the continuous assessment of risk on a forecast basis.

If you look through the literature and talk to those in industry, what impressed me were how completely two risk streams are addressed as different problems. Prices of commodities are monitored by finance groups and emission data by environment groups. The two seldom sit down and consider: What does our carbon compliance posture probably appear like next quarter, particularly when taking a note on the instructions where prices of raw material are heading? This dissertation aims to make a partial contribution to answering those two dialogues.

But machine learning can help with that, and change in fundamental data is one challenge that many aren't aware of. After big disruptions, things change in commodity markets. The existing regulations are amended. A training model that satisfies the pre-pandemic price behaviour may end up being a victim to the post-pandemic price environment although the technical implementation of the model is correct. This is the field at which the concept drift detection is targeting. It provides functionality to recognize if the base on which a deployed model is based has changed; this way, predictions will not become invalid. This was done by the development and integration of these methods into an integrated system for risk in industry, which was the main body of this project.

1.2 Problem Statement

The main question in this dissertation is simply expressed. There is no available, adaptive, integrated AI system today for industrial operators that dynamically tracks simultaneously commodity price regime risk, and carbon compliance risk, while remaining calibrated as market and regulatory conditions change. Each tool is unique and required for different tasks, plus there's no way to take account of new changes to the distribution – tools lose their accuracy over time and tracking is not possible.

In essence, this is leading industrial decision makers to base their sense of the price regime on gut feel, or outdated compliance estimates based on models built prior to the last regulatory change. It is an expensive error of judgment either way – to mishandle the procurement in expectation of a better price regime, or compliance fines for non-responding to compliance signals. This dissertation aims to answer the research question whether both the risk dimensions can be solved via a unified, drift-aware AI framework that is designed, developed and then empirically validated.

1.3 Objectives

Direction of the work is guided by 6 objectives. To actualize the three-part modular structure of architecture, which represents the first part – a commodity price regime predictor, carbon conformance risk classifier and carbon concept drift detector. Thirdly, to build and validate the price regime prediction model based on the historical data of the price of WTI crude oil and LME copper. Third, to develop a Carbon Compliance Risk Classifier based on features of emission intensity based on Indian Industrial benchmarks. Fourth, to improve ADWIN and Page-Hinkley drift detectors for continuous monitoring of the prediction streams, and for starting the recalibration whenever there is distribution changes of the data. Finally, to create an interactive Streamlit dashboard to display combined risk products, without having to ask a programmer to set up. Sixth, to use adaptive framework and static baselines for comparison in an appropriate manner to determine if there are truly performance improvements gained by using the adaptive design.

1.4 Scope and Limitations

The commodity price component is the price of two markets such as WTI crude oil and London Metal Exchange's LME copper. The carbon compliance module has been built on simulated data – there are no plant-level real industrial emission monitoring data in India and the simulation is calibrated with respect to BEE PAT scheme benchmarks – simulation does not have the capability to reflect the complexity at the plant level in this aspect.

Evaluation of the system not deployment, using single-machine, retrospective simulation. Again, not to beat around the bush on this — but as stated in 4, the performance metrics are for the framework in controlled circumstances — obviously we need something to constantly benchmark performance against as we go onward and need field validation for it.

1.5 Why This Matters

This work contributes in various aspects. But it also shows, in practice, that with an industrial risk intelligence system Python open-source tools are more than enough if you don't have enterprise-grade infrastructure – which is extremely critical for SMBs in India, who are unable to afford commercial analytics tools. To the best of my knowledge, concept drift detection in context of integrated industrial risk monitoring has not been widely used so far. In possibility space, the message that commodity price risk and carbon compliance risk go hand in hand - carbon policy decisions have a direct impact on the cost competitiveness of energy-intensive processes - has not yet been fully institutionalised in industry. It is in this framework that the link between the two worlds, the live and the video world, takes place.

1.6 Structure of This Dissertation

Chapter 2 discusses the literature on commodity price regime modelling, carbon compliance risk as well as concept drift detection and charts the space which is researched in the current work. The methodology, data acquisition, feature engineering, Dashboard construction, evaluation design and module design are presented in Chapter 3. Chapter 4

presents the results of the empirical study and discuss implications of the results. Chapter 5 shows the results, points of the limitations and future directions for the study.

Chapter - 2

Literature review

2.1 Why These Three Literatures?

In order to design the framework in this dissertation, I had to take on the task of engaging with the research traditions of both economists and data scientists when it comes to modelling commodity price regimes, as well as those of the machine learning community around carbon compliance risk prediction, and those of the concept drift detection literature for releasable algorithms. Then, a selective survey of the conclusions and discussions related to their work is presented, which most pertain to the design decisions made in Chapter 3.

2.2 Commodity Prices, Volatility, and the Regime Problem

The people who noticed a long-term monotonic trend in the prices of crude oil or copper may have called out, "Well, no, oil prices (or copper prices) certainly don't behave homogeneously over time. Long quiet episodes are followed by periods of violent movement which appear to be governed by a rather different statistical process. This intuition has been formalized and has brought commodity market researchers to work for some decades.

To this end we will focus here on some commodity time series which show distinct spans of high and low volatility, and in 1982 Engle introduced the GARCH model to model such time-varying conditional volatility. In principle GARCH and the extensions of GARCH explain within-regime volatility quite well, but they can be problematic explanations in that they explain step-changes to some extent in the statistical character of volatility between regimes. This was a difficulty in "Hamilton's 1989 Markov switching model where the regime jumps were assumed as a small number of hidden states with assigned series of transition probability. In 2005, Geman studied commodity derivatives; one of the few very complete studies of the structure of these markets compared to equity or fixed income markets – mean reversion to production costs, seasonality of the demand, and amplifying effects from the dynamics of the inventories.

The space of machine learning is joining in the space of regime with competitive results. Using high dimensional feature representations, Gaussian MM can find latent regimes without a parametric structure. In 2018, Fischer and Krauss showed how deep learning models can perform on a level par with classic approaches on benchmarks for financial predictions. The performance of Chen and Guestrin's 2016 XGBoost was good but it was applied to regime classification later with interesting results, though the running time was less when doing so using tabular-based features in financial data. The problem with all of these methods is that they are historical based and applied only statically - in the event that the market structure changes, they do worse without having a mechanism to identify that change or try to react accordingly.

2.3 Carbon Compliance Risk: From Reporting to Financial Management

This move – from carbon compliance as an optional informational obligation to seeing it as a financial risk – has been going on in Europe for roughly a decade and a half, and is now picking up in India. A key aspect of the EU ETS is that it created a market where the amount of allowed carbon emissions became a real-time, if albeit connected, economic variable: the "representation of a private currency", as Daniel Gerv was called, in the energy sector. An analysis of the mechanism by 2011 by Chevallier explained and showed their direct effects on competitiveness of energy-consuming industries.

While the PAT scheme does not impose a carbon price, there is a similarity in action to applying a price: For companies that are not at that energy level, there are similar financial penalties, and companies have to pay the same sort of certificate. Such a standard of performance has been reported in different regions of the world in the Sixth Assessment Report (2022) of the IPCC. The EU's CBAM adds a new context with its international scope: compliance to the CBAM will not just be localised, but will also be an issue of export competitiveness, offering a new impetus to the previous studies on PAT.

ML to predict compliance risks is a developing literature and not yet well explored. Random forest (RF) and gradient boosting classifiers (GBM) have been used to predict regulatory violations in these classifiers. LSTM networks have been used for modelling temporal dynamics in terms of sequences of emission intensities. One thing missing from this work is that as the process changes, equipment gets upgraded or if the compliance ranges change, there is no mechanism to update the models. If the classification takes place before the revision, a trained classifier will regularly do an underestimate and no one within the company will have any idea that things are amiss.

2.4 Concept Drift Detection: Foundation and Key Algorithms

Concept drift is defined as a shift in the joint probability distribution P(X, y) between training and deployment time – X are the input features and y is the prediction target. In layman's terms: The world that the model has been trained for is different from the one in which the model will be deployed. Sometimes drift is quick—such as when a regulator changes the regulation overnight—and sometimes it's slow, such as when something about a market changes in terms of volatility over months in response to a structural supply change.

The Gama et al. 2014 survey is a de facto reference of drift taxonomy, drift detection algorithm and adaptation strategies. Their key assertion, that detection and adaptation are two distinct issues, is a major factor forming the architecture described in Chapter 3, if nothing else because of their separation in the design process.

There are two algorithms that are key to this project. ADWIN, proposed by Bifet and Gavalda in 2007, maintains a window of recent observations of some size and periodically tests to see whether or not the distribution of observations in the more recent sub window is statistically distinct from the distribution of observations in the previous sub window. When the test plot exceeds the confidence level, drift is detected and the window is decreased to eliminate pre-drift data. The Page–Hinkley test, introduced by Page in 1954 for industrial quality control, is a helpful feature in a risk management context, where there may be a financial impact for a false positive, a false negative is detected by monitoring the cumulative deviation from the running mean of an observation sequence and issuing an alarm if the cumulative deviation exceeds a specified threshold. It works faster and is more sensitive to sudden mean changes as compared to ADWIN and is also a complementary algorithm to it. The River library is a clear and concise Python implementation of both algorithms in a single streaming machine learning framework as mentioned in the Paper by Montiel et al. 2021 in JMLR.

2.5 The Gap This Study Addresses

After reading over the three pieces of literature, I can see how it's a problem, and how in hindsight it's a bit of a surprise that it's not yet as much of a problem in such a practical use. Regime-modelling of commodity prices has a well developed arsenal of methodology. In this field containing Carbon compliance risk prediction, there is a literature that deals in an applied context. The theory for detecting concept drift and algorithms are well-developed. What is missing is an integrated solution that combines them all and a common adaptive layer.

The most similar study was that of Zhang, Liu and Wang (2020) which suggested multi-task learning for the forecasting of energy prices and emissions with a fixed model (without detection of drift). Compared to recent manufacturing risk dashboards that rely on machine learning with predictions, Kumar and Garg's latest 2021 guide relies on statistical process control. Drifting compliance risk was not linked to price in both papers. This integration gap is significant as in the event of an energy price regime entering a crisis, at the same time energy consuming processes will also be affected and their emission intensity will increase if replacement with low carbon energy sources is not possible on short notice and easy replacement terms. A common platform can monitor both dimensions and a common drift detection layer can be shared and identified potential compound risks which are unknown to the tools monitoring each dimension individually.

2.6 What the Literature Tells Us to Do

Three lessons from this review shaped the methodology in Chapter 3. First, GMM-based unsupervised regime identification followed by supervised classification has shown consistent ability to recover economically meaningful market states without labelled training data. Second, Random Forest remains a strong choice for tabular compliance data with mixed feature types; its well-calibrated probability outputs suit threshold-based risk alerting better than deep learning. Third, deploying ADWIN and Page-Hinkley in parallel provides better coverage across the gradual and sudden shift types that both appear in industrial data environments. These three choices define the core of the methodology that follows.

2.7 The Indian Aerospace Manufacturing Context and Emission Monitoring

The academic literature on carbon compliance risk has been shaped primarily by the European experience of the EU ETS and by research on large utility and heavy industry sectors. There is comparatively little published work that addresses the specific compliance risk characteristics of precision aerospace manufacturing in the Indian context, where the relevant regulatory regime is the PAT scheme rather than a carbon market, and where emission intensity is driven by the energy consumption of CNC machining, surface treatment, and environmental control systems rather than by combustion of fossil fuels in the way that characterises utility or steel sector emissions.

This literature gap has a direct impact to the design of the compliance module in this concept study. The synthetic data used for the study is for the steel industry and, therefore, for a full-up implementation, the value of emission intensity and also the value of the regulatory threshold value would need to be re-calibrated for the purpose of applying it to the steel manufacturing industry under the PAT scheme. As mentioned in chapter 3, this is also a limitation mentioned and suggested as future research direction in chapter 5.

The general trend, though, remains: As Indian aerospace producers increasingly ramp up and face rising pressure from offshoring customers that are subject to compliance intelligence demands of their own, the need for a stronger, more future-thinking overall approach in India's aerospace industry will intensify. The framework developed in this dissertation will be able to satisfy a need when calibration data required are available.

2.8 Commodity Price Exposure in Aerospace Manufacturing

Where commodity price exposure is important, however, the difference lies that an aerospace precision manufacturer like CIM Tools, for example, will be exposed to commodity price in a significantly different way than an energy commodity exuded by an industrial manufacturer, say in power generation or petrochemical production. The primary raw material for aerospace machined parts is aluminum alloys, Titanium alloys, Nickel based alloys and high strength steel. The most important of these on a typical volume basis are aluminium and titanium and both are quoted on exchange traded price series over many decades and are predominantly used in the manufacture of commercially aerospace products.

Its price is determined by London Metal Exchange (LME) contracts and influenced by smelter capacity, export and production trends in China, energy prices that are used for melting it and absorption trends of key industries that use the metal, like the automotive, construction and aerospace industry. Much less traded as a traded commodity but the price in the spot market is traded, and the prices include geopolitical factors, including activities in the Russian, Ukrainian and Chinese mines, which came to the fore in 2022 when the supply chain was interrupted. These are an aerospace component manufacturer's key margin pressure items on current contracts and uncertainty on new programme bidding, as the continual upward regime transitions in these materials result in a reduction in the aerospace makers' margins.

Among commodities, the price of titanium is a proxy in the price regime prediction module created in this dissertation: The price series are long enough to be useful in the module construction, a requirement driven by the need to have a long enough historical period to observe any trend.For the price prediction option in the price regime prediction module developed in this dissertation, the price series of WTI crude oil and LME copper has been used: All due to the fact that prices of these commodities are available for long enough historical prices horizons needed to observe a long enough trend for the prediction module construction. However, the proposed methodology (regime identification + XGBoost + ADWIN) is easy and directly transferable to aluminium or titanium prices series, with a proper calibration. The transferability is part of the frame-work design philosophy that takes on a modular approach.

Chapter – 3
Methodolgy

3.1 Research Approach

The design of this project is based on the design science research methodology which includes building the artefact, setting it up in a field of practice, and describing the behaviour of the artefact in context. The overpowering importance is the work itself; the framework; the empirical evaluation is to establish the validity and usefulness of the work. Data acquisition and processing, construction of individual modules, system integration dashboard building, calibration, and comparative static baseline between.

3.2 Data Sources and Preprocessing

The price regime module uses all the price data of the daily trading price of the WTI crude oil and sales data of the LME copper from January 2000 to December 2023. WTI crude prices symbolise the energy cost component, which is the biggest non-fixed cost of many industrial processes. Copper is a good base metals proxy, and as metal demand is very closely related to industrial demand it is also a good forward looking indicator of global industrial demand. Using the public API of the U.S. Energy Information Administration (EIA) to retrieve data for WTI and the Quandl API for commodity data to retrieve data for copper prices. This 23-year window captures the largest set of structural periods to include a supercycle from 2003 to 2007, a financial crisis in 2008, an oil price collapse from 2014 to 2016 and a COVID-led demand shock in 2020 and energy market dislocations in 2022. This diversity is deliberate, for it is only when a framework is supposed to include regime changes that it's employed for a history in which regime changes occur.

The missing observations were forward filled which arose because of the various holidays schedules used using the last available Price. To achieve variance stability through different price level regimes these all series had to be combined using log transformation.

The carbon compliance dataset was created from scratch, however, as no data on the actual emissions exists, measured on the plant-level in India. Going back to something I know a bit about, (the integrated steel sector), I made up some data of 5000 observations of integrated steel companies over a period. The baseline emission intensity was randomly drawn from a Normal distribution with the corresponding published BEE sector benchmark mean of 2.1 tCO_2 equivalent per tonne of output and a standard deviation of 0.35. The regulatory threshold was 2.5 tCO_2e/tonne for the first 3,799 observations and 2.2 tCO_2e/ tonne for the remaining observations which simulates decreasing the regulatory threshold in successive PAT cycles. A second drift was attempted at 2500obs with a downward drift of 0.25 units in mean emission intensity, that is, improved process efficiency. The compliance labels are dichotomous – either non-compliant if the simulated intensity is above the threshold or compliant if below the threshold. This will result in an actual rate of non-compliance of ~20 percent throughout the data set.

3.3 Feature Engineering

In the price regime module features contain dynamics in a range of time scales at the same time. Log-returns are Daily Moment statements. The three moving averages (5 day, 21 day and 63 day) offer a trend-like view at weekly, monthly and quarterly time periods, respectively. In order to estimate the volatility for each scale, rolling standard deviations are calculated at the same horizons. Previous financial machine learning research has uncovered that three technical indicators, the Relative Strength Index (RSI), the Bollinger Band width (BBW) and the Macd signal (MACD), offer information for regime forecasting in addition to the raw machine learning signal (return). As the movements in both markets (WTI/crude and Copper/Metal prices) tend to signal and not commodity specific price shift, the cross-correlation of the last 21 days rolling values were used as macro-economic context features. Only the first components of the full feature matrix were retained (95 per cent of the cumulative variance) to train the model.

For the compliance module, the most important engineered features capture proximity to the regulatory boundary rather than absolute emission levels. The rolling 30-day emission intensity trend measures directional movement toward or away from the threshold. A threshold proximity score, defined as the difference between current intensity and the applicable threshold in standard deviation units, gives the model a direct representation of the margin to the compliance boundary. An abatement effectiveness ratio tracks installed abatement capacity utilisation. An energy mix carbon intensity index weights the carbon content of fuel inputs by consumption share. Categorical features, production process type and regulatory jurisdiction tier, were target-encoded to preserve compliance rate information.

3.4 Module One: Commodity Price Regime Prediction

The price regime module is a two-stage pipeline. Stage one is unsupervised: a Gaussian mixture model with three components is fitted to the PCA-compressed feature matrix. Component count was chosen by fitting GMMs from two to six components and selecting the specification minimising the Bayesian information criterion. Three components consistently produced the lowest BIC across initialisation seeds, and the resulting regime assignments align with the qualitatively distinct market periods visible in the historical data, which I treat as a useful interpretability check.

Regime labels from the GMM become the supervised target for stage two: an XGBoost gradient-boosted classifier trained on a rolling 252-day window. The model generates probabilistic regime predictions for the following 21 trading days. Hyperparameters were tuned via 5-fold time-series cross-validation, optimising macro-averaged F1 score to account for the class imbalance across the three regime categories.

3.5 Module Two: Carbon Compliance Risk Classification

The compliance module is a Random Forest classifier with 200 trees and maximum depth 12. Random Forest was chosen over XGBoost here partly to introduce model diversity into the integrated system, and partly because its averaging over multiple trees produces well-calibrated probability estimates important for the dashboard’s threshold-based alerting. SMOTE was applied to the training split to address the 20 percent non-compliance rate without contaminating the test evaluation. The held-out test set comprises the final 20 percent of observations, stratified by compliance label. Evaluation metrics include AUC-ROC, non-compliant class F1 score, and expected calibration error.

3.6 Module Three: Concept Drift Detection

Both of the predictive modules will be surrounded by a monitoring system (drift detection layer). Monitoring is done based on the price regime prediction error stream generated by ADWIN for the confidence parameter, delta equal to 0.002, that is, the error rate is monitored whether it changes in a manner that is outside of the range of random fluctuations or not. As regulatory changes are anticipated to lead to an abrupt change in compliance, the Page-Hinkley test is used here with lambda = 50 and forgetting factor = 0.005, both of which are more sensitive to an abrupt change in the mean, and hence more suitable for the context of compliance.

A detector signal which has drifted will be re-trained as follows: For the price regime module the XGBoost process will be retrained with the prior 252 trading days; and for the compliance detections module, the Random Forest process will be retrained with the last 1000 observations. Either protocol processes are automatic. The strong record of drift events and results from the recalibration are stored in a structured record, which is supplied to the drift monitoring panel of the dashboard. Each of the 2 detectors sends 10 observations per minute through River's streaming API: that is, the full set of observations is not loaded into memory, but rather processed incrementally.

3.7 Streamlit Dashboard

The Streamlit Industrial Risk Dashboard ensures the framework's outputs are accessible to a non-programming user. The front end is run by the model code (also known as Python-native), without any translation layer. The interface contains 3 panels: (1) price regime panel, which shows the probability of the current regime and historical regimes timeline; (2) compliance risk panel, which shows the probability of being out of compliance, a colour coded level of risk indicator and a 30 days threshold proximity indicator panel; and (3) drift monitoring panel depicting the timeline of all the detected drift signales, the detector that triggered it and a pre and post recalibration performance comparison. Interactive controls let the user adjust the historical window, switch between commodities and adjust levels of alerts.

3.8 Evaluation Design

The two modules use static baselines for comparison with the adaptive framework. The price regime module makes up one static XGBoost trained on the first 252 days and is kept untrained and one naive persistence forecasting regime persistence. The module for compliance testing: A static Random Forest (not recalibrated) was built on the first 80 percent of the data, and applied to the final 20 percent. The period of evaluation for price regimes is 2022-2023 and it involves the crisis of the energy market, that was caused by the War in Ukraine. The whole synthetic dataset (2 drift events) is taken into consideration within the compliance evaluation. The considered performance metrics are the classification accuracy, the macro-averaged F1 and the AUC-ROC, and the detection delay, false positive rate and post-recalibration improvement of drift detection. Any experiments are performed in Python 3.10 with fixed random seed.

3.9 Data Quality Considerations and Preprocessing Decisions

The pre-processing options merit further explanation of some of these options than is provided in Section 3.2. Taking commodity prices at this stage, before feature engineering, is done in the spirit of empirical regularity, the volatility of commodity prices is comparable to the level of prices, hence the raw commodity prices are heteroscedastic, which violates the statistical assumption of many of the modeling methods. This multiplicative price dynamics is transformed (after log) into additive return dynamics, which is easier to deal with in the case of the GMM regime identification and with the XGBoost classifier.

We decided on these two very specific commodities (WTI oil and LME copper) rather than a more broad commodity basket because we think that our modelling philosophy of "some well-chosen commodities with long histories and good price discovery would be better than more commodities with less price discovery or with shorter histories" would be applicable. Backing the daily price series of WTI crude is far more information on prices for this commodity than is found for most industrial commodities, and the daily price series has a continuous record dating back to the mid-1980s (with limited data available earlier). The copper being traded on LME is exposure gradually compliment to the demand dynamics for industrials. Realizing the practical accessibility goal of the framework, both of the series now have free public APIs.

The forward-filling of missing holiday observations deserves a brief methodological note. In financial time series analysis, the appropriate treatment of non-trading days is context-dependent. Where the purpose is to model continuous underlying price dynamics, forward-filling is standard. Where the purpose is event-study analysis or high-frequency modelling, gaps should be handled differently. For the regime identification task in this dissertation, where the relevant time scale is weeks to months rather than days, forward-filling introduces no material distortion.

3.10 Model Architecture Rationale: Why Not Deep Learning?

A question that arises naturally in any contemporary machine learning dissertation is why the predictive modules use gradient boosted trees and random forests rather than the deep learning architectures, particularly LSTM networks and transformer-based models, that have attracted significant research attention in financial and industrial prediction tasks. The answer involves both principled methodological reasoning and honest acknowledgement of practical constraints.

On the methodological side, the tabular feature representations used in both predictive modules, engineered rolling statistics, technical indicators, and emission intensity ratios, are the type of structured input for which gradient boosted trees have consistently demonstrated competitive or superior performance relative to deep learning in the academic literature. Deep learning architectures offer particular advantages when the input data has high-dimensional sequential structure, as in raw time series or text, but for moderate-dimensional tabular features the ensemble tree methods tend to require less data, train faster, generalise better, and produce more interpretable outputs. Interpretability is specifically important for the compliance risk classifier, where risk managers need to understand why the model is predicting non-compliance in order to take corrective action.

On the practical side, deep learning models require substantially more computational resources for training and hyperparameter optimisation, have more sensitive calibration requirements, and are considerably harder to retrain in the incremental recalibration protocols that the drift detection layer demands. A Random Forest that retrains on 1,000 observations in under a second is genuinely practical for an automated drift-response recalibration cycle; an LSTM that requires minutes of GPU time for fine-tuning is not, at least not without infrastructure investment that would price the framework out of the accessible deployment context it targets.

The open question for future work is whether incorporating sequence-aware architectures, possibly through a hybrid design where a transformer model generates sequence embeddings that feed into an XGBoost classifier, could improve performance enough to justify the additional complexity. The results in Chapter 4 suggest that the current architecture is already achieving strong performance, which raises the bar for any more complex alternative to clear.

3.11 Dashboard Architecture and Deployment Considerations

With the benefits of the architecture that have been well documented in the Industrial Risk Dashboard and using that architecture for the Streamlit deployment, there are some practical implications that warrant a discussion. The Python-native way of Streamlit is extremely helpful when we need to rapid prototype as well as when small groups of technically knowledgeable customers need to consume analytical-ish functions. However, there have been known limitations on production deployments at scale: First, Streamlit's execution model is to re-run the entire block of application code upon each user action, and as these models get larger and these data sets get bigger, the compute may be a cost concern. Second, the lack of feature-rich session state management in Streamlit compared to Web application frameworks is a limitation.

Taking all these things into consideration, Streamlit is a perfect fit for the purpose of this dissertation with the aim of demonstrating the feasibility and technicality and to build a working prototype. If your deployment is to an industrial environment, requires several people to use the same system at the same time, and needs to be integrated into real-time data feeds, something more substantial would likely be required, including separation of the model inference layer from the presentation layer, a backend that uses an API, and some kind of framework on the frontend that has a better system of handling sessions. These are adjustment to the system of engineering and not a major design change and can be achieved with the detached construction of the framework.

The scale-up into operations would present the greatest technical challenge with regard to the use of real-time data feeds. Current simulation system can only provide data to dashboard in “batch” – no realtime monitoring. The real-life potential industrial use case should allow for the dashboard to be fed actual data feeds from emission monitors, production planners, and even commodity price feeders. While there is a clear relationship between the API's drift detection layer and a real-time data feed, you will need to think through a lot of things in production, such as data quality, latency and error handling, especially, the API from the River project is optimized for this use case.

Chapter – 4
Discussion

4.1 A Note on Reading These Results

All numbers mentioned in this chapter are from a good simulation and do not reflect an actual deployment in industry. Not only are they fake, but they actually backfit and measure the output of real models on out of sample data. But they are results of simulations, and I think I have to state what that means, interpretation-wise. I'm giving them as a sample of what can be achieved with the framework under very specific conditions – this is the initial step – this before field validation.

4.2 Price Regime Identification

The GMM wanted to select 3 regimes and I liked the interpretations of these regimes very much. This is the low volatility positive momentum regime (roughly 42 percent) on all trading days in the entire sample, which occurred during the 2003-2007 supercycle and the 2016-2019 recovery period. Regime 2 which contains about 35 per cent of the observations captures the high-volatility behaviour and the mean reversion of uncertainty and transition. At 23 percent, Regime 3 is concentrated around occurrences of the crisis and correction events: the days surrounding the 2008 Lehman bankruptcy, the 2015-16 oil glut and the March 2020 COVID demand shock are all heavily loaded with Regime 3. It is very convenient to have an algorithm that had no information about former occurrences and could still recover the structurally meaningful periods, since it is an entirely unsupervised algorithm that is used for regime identification.

4.3 Price Regime Classifier Performance

On the held out 2022-2023 evaluation window, the macro-averaged F1 score of the adaptive XGBoost classifier was 0.81. The static baseline, trained only at the beginning of 2022 and not updated ever has 0.67. The majority of degradation was due to Regime 3 misclassification, due to the unusual volatility structure observed in energy markets in Europe in the second half of 2022. The score on naive persistence baseline was 0.52, which isn't too difficult and by no means is it a small amount of signal, that's provided by machine learning beyond inertia.

There is a very tangible and important narrative that is 1) regime transition 2) alerts. At the evaluation time, the adaptive model has yielded 18 alerts, of which 14 were later confirmed as a regime change with 21 day time horizon giving a precision of 0.78 and a recall of 0.82. This led to just 12 alerts for the static baseline of which 7 were correct (precision=0.58, recall=0.54). Thus, while the static model is less accurate — it is also quieter when it is needed most, which can be the more dangerous mode of operational risk failure.

4.4 Carbon Compliance Classification Performance

The test set AUC-ROC for the adaptive Random Forest is 0.89 as compared to the static baseline (AUC-ROC 0.77). The class F1 score, which is the one that counts operationally–with false negatives meaning a regulatory cost–was 0.81 for the adaptive model and 0.62 for the static baseline. 19 points is no chump change.

Feature importance outputs are given to make inferences. These three variables (rolling 30-day emission intensity trend, proximity score to thresholds, energy mix carbon intensity index) account for 64 percent of the decision weight in the model. If the plant has been steadily ratcheting up its intensity use for a month and is rapidly approaching the compliance limit there is no comparison to the risk of compliance to a plant that has been well below regulatory limits on intensity use but may have a long history of high intensity. What is important is that a basic compliance dashboard which monitors only these three criteria would be able to capture a large amount of the compliance risk signal.

The numbers that I find most interesting are from the drift experiment. A fairly steep decline then occurred in the AUC-ROC of the static model between observations 2,500 and 2,700 (from 0.79 at the first 2,500 observations to 0.63 at the last 2,500 observations) after the synthetic drift at observation 2,500. At observation 2,547 (47-observation delay), however, the adaptive model sensed its limitation, made the change and then tracked its value to within 100 post-recalibration observations of the value 0.85, when the second drift of the detector alerted the Page-Hinkley detector, which recalibrated to give AUC-ROC 0.88. The second case: the sudden mean shift means the PH relaxes to it, hence preempting the notification of change in a PH; that's why this one gets detected ‘faster'.

4.5 Concept Drift Detection: What the Two Algorithms Each Bring

At first, I felt that it was too much a good thing to run ADWIN and Page-Hinkley at the same time, but it does! The price regime module of ADWIN detected the pricing change of the energy market in 2022 in only 68 trading days. A threshold based alert (with another comparison) was 142 days behind in attaining the same level of confidence. In the compliance module, PH was able to detect the abrupt switch up of the threshold in 31 instances while ADWIN detected this in 89 instances. This complementarity isn't hypothetical; ADWIN is great for slowly moving streams of data, PH is great for very sharp jumps of data and "real industrial data streams" do both of the above. With the ADWIN resulting in a rate of 2,3 per cent and the PH in a rate of 4,1 per cent, all the false positive rates were below the acceptable operational tolerances throughout the evaluation period.

4.6 Dashboard and Usability

In this instance an onlooker (five postgraduate data science learners) watches through a data science lens, and an informal usability test was conducted with five postgraduate data science learners, but no construction. The average page render time was 1.8 seconds, which is lower than what a customer will consider tools to be responsive. Three panel format was user friendly at initial use and for all evaluators. Four out of five reported that having both the Pricing regime and compliance risk & Drift monitoring in a single view provided you insights you wouldn't get by individually viewing each of the two panels and could impact your decision making. One evaluator commented that the alert level control may not be properly labeled - I will do that. The most attention was garnered by the drift monitoring panel; some evaluators were not familiar with it, and needed some assistance to translate the information into the right context, but onboarding documentation would be required in an operational deployment.

4.7 The Bigger Picture

The key gains are 21 % Macro-averaged F1 Score as opposed to Macro-averaged F1 Score of Price Regime and 12 % AUC-ROC as opposed to AUC-ROC of Compliance when using Static framework versus Adaptive framework respectively. But what is more important is when those improvements will be made. The two models of adaptive and smooth stationary models have the same performance when the system is in a stable state. The benefit is heavily on the windows between drift events, where risk assessments are not very different, right or wrong, and models are more static and least likely to be wrong. Obviously, the adaptive architecture wouldn't be nearly as useful, relatively, for a business enterprise that is operating inside a industry which is, by itself, stable, with no modifications in legislation. However for any longer time frame, one like that is historically rare and it is in those more turbulent time frames that the vindication of the adaptive (agile) lens exists; in those time frames it is when risk management can be challenging at best.

4.8 Sensitivity Analysis: Drift Detector Hyperparameter Effects

The behaviour of the drift detection layer depends on the hyperparameters setting for ADWIN, as well as the hyperparameters setting for the Page Hinkley test; optimal selection of those hyperparameters requires enough studies and analysis to gain an insight into how the delay and false positive rate trade-off will behave. All the major hyperparameters have been grid searched systematically while developing; there are some interesting results that I wish to peruse a little bit.

The most important hyperparameter of ADWIN is the statistical threshold (delta confidence parameter) for determining a drift. A smaller value of delta will result in higher evidence of drift being required before drift is declared, a lower ratio of false positives and higher latency to detect drift. The context was the price regime and the values used were the 'Ecos' with details of Delta: 0.001-0.01. For energy market transition 2022, the false positive rate at delta = 0.001 was 0.8 percent, the detection delay was raised from 68 days at delta = 0.002 to 94 trading days. Days to detection were decreased to 47 and the false positive rate was raised to 4.1 percent when delta was lowered to 0.01. The number chosen in this paper, 0.002, is thus an operationally defensible balance: in a context with a procurement risk, 2-3 months of playtime for a major regime change is enough; conversely, with a false alarm rate of <2.5 %, detection cycles based on false alarms lead to using precious resources for unnecessary recalibration.

For the Page Hinkley test, the value of the threshold lambda parameter is more directly related to sensitivity to mean changes of varying magnitudes. At lambda = 25, the detector flagged too many small changes in the distribution, which were not indicative of a change at the regulatory level, giving rise to a false positive rate of 8.7 percent. The detector identified a more significant compliance dataset mean-shift (the process degradation) at lambda = 100 but the smaller synthetic drift event (the process efficiency improvement, producing a more gradual mean-shift) was not detected until after a delay of 180 or more observations. Both synthetic drift events were detected at the operationally useful setting delays by using the setting with $\lambda = 50$ at a false positive rate of less than 5 percent. This parameter should be adjusted for future deployments to the regulatory revision magnitude that is associated with the applicable regulatory regime.

4.9 Feature Importance Deep Dive: Implications for Simplified Monitoring

The results of feature importance derived from the compliance risk classifier dwarfed the "technical" performance results. The problem is that since these three (rolling emission intensity trend, threshold proximity score and energy mix carbon intensity index) are the three dynamic attributes that have been identified as being dominant over longer-term operational attributes then whether it makes sense to have a simpler rule-based monitoring system that only tracks these three attributes and achieves a useful fraction of the full classifier's performance?

I came up with this simple heuristic to use for this test: a tendency of the rolling emission intensity 30-day trend of pointing upwards and a proximity below 1.5 SD with the threshold score means that it is non-compliant-risk. This rule was much less accurate on the test set, with an AUC-ROC value of only 0.71 and an F1 score of non-compliant items of only 0.58 compared to the F1 score of the full classifier of 0.89 and non-compliant F1 score of 0.81. This margin has been considerable, especially with regard to the F1 metric which is most critical operationally, and it shows that there is indeed an added value in the whole feature set and the non-linear modelling of the Random Forest method as compared to a simple threshold rule. Not that the heuristic rule doesn't perform badly either, but for any operator that lacks the technical expertise to establish and resources to maintain a machine learning pipeline, a well optimized dashboard of three indicators to track compliance' intelligence for the future will greatly outperform.

This reveals some of the comments in practical implications – as was specified there, it is not the most sophisticated model that is the most important – it is to start some monitoring which is necessary by most industrial operators and to do it in real time. Predictive with prescriptive translates to maximum performance of the data used by computer science experts with the same approach of this dissertation, and minimum performance by “digitally savvy” operations with the same approach of heuristic in this dissertation but with only three indicators.

4.10 Cross-Module Interactions: When Price Regimes Predict Compliance Risk

A key selling point for an integrated framework over two would be that there would be a strong connection between the commodity price regime and the carbon compliance risk. But, from empirical results there is some evidence that substantiates this relationship, in simulation it is indirect.

The cost pressure is linked to episodes of regime change in the price regime score (Regime 3 ‘energy price crisis') when the synthetic compliance dataset imbued higher energy intensities in which a change of production mix would appear to be most likely to have taken place. The equipment might also operate more aggressively in the real life scenario when the asset replacement is not economical during the cost pressure period or the fuel input into the process may be changed to more carbon fuel input when the relative price of other forms of fuel input is increased.

The direct emission intensity features, also coupled with the price regime state and compliance risk level simulation, highlighted the moderate but detectable relationship of the two; observations, associated with regime 3, were 34 percent more likely to be associated with the compliance risk level state "elevated" when compared to observations associated with regime 1. This correlation was not strong enough to suggest the use of the price regime signal as a direct input to the compliance classifier in the existing design of this system, however using the estimated likelihood a regime for cross module feature engineering (utilizing the price regime signal as input to the compliance risk model) may be of interest for future versions. This would truly be the integrated framework concept, as opposed to just co-existing on the dashboard: price regime module feeds into the compliance module.

Chapter-5

Conclusion

5.1 What This Work Set Out to Do and What It Did

This dissertation started with a practical observation: two exasperating problems of Indian industrial operators, the instability of commodity price regimes and the carbon-compliance exposure, are almost always discussed as distinct issues by various teams using dissimilar tools. This was done to come up with a working model that could address them both over one adaptive platform, be empirically demonstrated that the adaptive design would do better than the alternatives of the day that were both static and needed enterprise-grade infrastructure but not open-source tools that could be brought together on a single platform.

The goals are met in the framework of the Integrated AI Framework of Industrial Risk Management created here. A running Streamlit application is a GMM-plus-XGBoost price regime model, a Random Forest compliance risk detector, and an ADWIN and Page-Hinkley drift component. The adaptive framework has a 21 percentage-point macro-averaged F1 improvement over static baselines the price regime task and a 12 percentage-point AUC-ROC improvement over the compliance task with the improvement concentrated in the period of drift events when the cost of risk assessment error is greatest.

5.2 Contributions

The donations are on three levels. I believe that the integrated industrial risk monitoring concept drift detection is a new contribution, technically, to my knowledge. The previous drift detection works have been performed under conditions of the benchmark classification task and one-time prediction problem, and not with the simultaneous risk monitoring of the commodity price regimes and carbon compliance. The particular design decisions such as the choice and calibration of the detectors or the design of recalibration procedures are not discussed in the existing literature.

The model empowers the operationalization of the structural relationship among commodity price risk and carbon compliance risk, on the conceptual level. As the energy prices are moved to crisis regime simultaneously energy intensive activities are thrown into a more costly situation and in the unlikely event there are still no cheaply available low-carbon equivalents that can be made available at short notice, increased levels of emissions. Having both dimensions monitored on a common platform indicates the situation of risk of compounds that would not be detected with the help of independent tools.

In practice, it is critical to show that it is possible to build such using River, XGBoost, scikit-learn, and Streamlit on a general-purpose machine since it is more readily available. Majority of the published industrial AI systems presuppose huge proprietary data and expert engineering assistance. This framework was developed by one undergraduate student using the assistance of the publicly available data and free available tools. One of the sides that render the contribution meaningful is the implication that the implementation of something of this scale makes sense to Indian mid-sized manufacturers with no large analytics teams.

5.3 What Practitioners Should Take Away

Any practitioner who may be contemplating this type of a system should highlight three points. Early and dynamic indicators of compliance, the trend of rolling emission intensity and the distance to the regulatory threshold are discovered to predict most of the compliance risk: about two-thirds. Tracking of these does not need a complex model. You ought to watch them in real time. Second, it is possible to forecast transitions of a price regime of a commodity with a reasonable accuracy of 2-3 weeks ahead based on published market data. Now, practically, we are able to state, by constituting even a simple capacity of the regime-monitoring in procurement planning. Third, any predictive model would stagnate over time as the markets and regulatory changes. Neither is concept drift detection an added value it boasts about. Stability of operation, over a time span extending beyond several months, is a prerequisite.

5.4 Limitations I Take Seriously

On synthetic data, the carbon compliance module is tested. The simulation is empirically adapted, the drift events are real life like events, but no simulated data captures exactly the autocorrelation structure, equipment specific variances, and measurement noise and other anomalies of the actual plant level emission monitoring. The performance numbers must be perceived as fair, and not as caused to work when compliance annex (with the real records of BEE PAT schemes) must be pleased only with a data-sharing setup with the BEE or an engaging firm.

The usability test included five postgraduate students as opposed to having the experienced industrial risk managers. Even the interface choices that a data science learner would consider more logical to use can be literally upon a baffling array to the plant operations manager who has never seen a probabilistic presentation of risk. The two commodity markets are also the two only markets that have been covered in the framework; it would have necessitated a re-assessment of both regime identification methodology and choice of feature engineering to make it to coal, natural gas, or even the agricultural inputs.

5.5 Directions for Future Research

The most pressing subsequent task is to have access to actual compliance data. Experimental partnership between the BEE and a member of the PAT-schemes industry with which we could acquire actual emission monitoring information would allow compliance module validation in the real-life setting and would reveal relevant refinements that would not be possible with synthetic data. The second priority is to implement the architecture of drift detection in federation where data is distributed in a chain of facilities and cannot be centralised. Detection of federated drift would render the framework practically relevant to operation of multi-site industry.

Since we have an additional type of input to the price regime model, which is the sentiment analysis of the commodity markets; we have a model, say FinBERT, on the news and regulatory announcements, which we would add to the price regime model to ensure a lesser detection lag and better transitions alerts because we would be adding forward-looking information that is not captured by the historical price patterns. Lastly, causal inference techniques fitted to post-drift data might tell a change in regulatory threshold but not a process technology change as the cause of an observed distributional change, and offer diagnostic information relevant to decision making and not a signal indicating a change of direction.

5.6 Closing Reflection

Something seems right to have a dissertation on industrial risk monitoring as having been inspired in the first place by observing the way price volatility influenced a business on fabrication at Jamshedpur. I have not commenced with an entire framework. I began with both real and consequential question in order to outline how I would obtain the means to answer these questions. The body of literature had to do with regime switching and concept drift detection and compliance classification. Engineering work helped such methods to become a working system. The testing ensured that the system being tested performs as desired under the conditions under which it was programmed to operate.

It is not necessarily the number of particular performance that I learn as a result of doing this project. The assumption behind this is that not every most technically ambitious AIs is intended to become the most helpful in the highest stakes industrial settings. It is them which a practitioner can comprehend, trust, and rely on as things evolve. The most crucial yet least significant component of the framework is the drift detection layer: not because it offers performance gains but because it is the system mechanism that causes the system to be truthful over time, signalling the point where the models have started to drift beyond those it is meant to model. To create such accountability to a system that will in turn be utilized to make consequential decisions is to give to a system a feeling of the correct engineering philosophy, and I am happy to close this dissertation with this note.

5.7 Implications for the Motherson Aerospace/CIM Tools Context

Practical application provided in Motherson Aerospace/ CIM Tools that led to the development of the present research leaves one wondering how the incorporated structure that I developed in the context of this dissertation would be utilized in the latter context. It is valuable to be concrete about this, as what can be of use about the risk intelligence framework it relies on, not just its technical effectiveness, but also, the specificity of the organisational/data environment, in which it would be employed.

CIM Tools run on site in Bengaluru four special aerospace manufacture factories and each factory delivers constant energy consumption, amount produced and quality of processes pulses of information. The current facility level has an accumulation of emission monitoring data (required by the BEE PAT scheme reporting requirements). The gap is between the collected data turned into a real time compliance risk indicator that may be availed to the production planning and operations management personnel in the form that would permit the operation management staff to take proactive decision. This dissertation has created a translation layer in the form of its Streamlit dashboard framework, and the calibration of the compliance risk module to PAT scheme sector requirements is such that adaptation to the specific regulatory requirements of the CIM Tools was a question of parameters re-calibration not a framework re-design.

The most closely positioned exposure in terms of priority as a material cost by CIM Tools on the commodity price side is the aluminium, titanium and specialty alloy inputs as opposed to WTI crude or LME copper. To adapt the price regime module to the series of these commodities would be to assemble the right history price information, in virtual reality a societal and business database, and re-estimate the GMM regime recognition, this time on the new series. Two-stage GMM-plus-XGBoost architecture would be straightforwardly reusable regime labels Correspondingly, the parameters of feature engineering would need to be validated on the corresponding historical episodes to aerospace material markets.

On a bigger level, a combination of the two capabilities would most help the integration of this framework into the work of CIM Tools, namely, the structured data infrastructure, which can be offered by the manufacturing excellence programmes of the Motherson Group, and the analytical capability of the framework. Years of experience in the precision automotive manufacturing industry have resulted in Motherson having highly developed systems of production data collection, quality monitoring and supply chain visibility. Mimicking these flows of information to the sort of integrated and adaptive risk intelligence system put in place here would be a truly differentiated ability to manage risks in the Indian aerospace supply chain - which, as the financial obligations mandate part of the CBAM filter through between 2026 and 2050, can become a real source of competitive differentiation in the export market setting.

5.8 Broader Lessons for Indian Industrial AI Adoption

The process of creating, constructing, assessing this framework, paired with the background of having a first-hand experience of how risk management really works in an industrial environment creates some observations regarding the state of AI adoption in the Indian manufacturing sector that I would like to record.

The first one is that the data infrastructure to enable heavy serious predictive analytics in Indian manufacturing is still more advanced than analytics practise that is based upon it. Most big manufacturing plants in India are generating huge volume of production, quality and energy data due to their automation systems. Not gathering data is what kicks off the performance, but integrating information: consolidating flows of information between different systems, of different frequency, of different quality levels and making common analytical territory. This is an organisational and systems architecture problem, not a machine learning problem, and requires more focus on the part of practitioners than the underlying machine learning models.

The second one is regarding the importance of interpretability. A model the outcomes of which can be justified by a risk manager to a regulatory auditor is better in the compliance risk context than a model with a marginally higher AUC-ROC but be unexplainable. The fact that the Random Forest produced feature importance results significantly showing that the agent of change in emissions intensity and proximity to the threshold that so heavily figures in the near future are the predictors of paramount importance are precisely the kind of explainable narrative a conversation about compliance is to require. There is nothing in making interpretable models preferred over marginally better black-box models which amounts to compromise with methodological conservatism; rather it is merely a practical necessity in order to be able to use it operationally.

The third fact and, perhaps, the most important to the greater problem of implementing AI in Indian industries, is that the concept drift detection layer in this system is not a purely technical feature. It is a statement concerning proper epistemic relationship between a prediction system and the organisation, in which it is applied. A system which assumes that it will work indefinitely is merely laying down empty presuppositions of false safety which will one day lead to the miscalculation of the judgments it will be called on to practice. A living system that constantly checks whether its models are still valid and causes recalibration in cases of failure is truthful with regard to what the system can and cannot know. There is no ethical requirement or poor practice in developing that honesty into the industrial AI systems; they, being systems will support consequential decisions made relating to regulatory compliance, procurement strategy and operational planning, and must do so.

Chapter-6

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Chapter-7

Annexure

INTERNSHIP REFLECTION — MOTHERSON AEROSPACE/ CIM TOOLS PRIVATE LIMITED

1 Overview of the Internship

This appendix provides a structured reflection on the internship conducted at Motherson Aerospace/CIM Tools Private Limited in Bengaluru, which formed an important part of the experiential foundation for this dissertation. The internship was undertaken in the domain of data analytics and industrial emission monitoring, and it offered direct exposure to the operational realities of an aerospace precision manufacturing environment that significantly shaped the research questions addressed in the main body of this work.

CIM Tools Private Limited, now operating as part of the Samvardhana Motherson Group following the 2022 acquisition, is headquartered at the KIADB Industrial Area SEZ Aerospace Park in Devanahalli, Bengaluru. The company supplies precision machined parts and sub-assemblies to commercial aerospace OEMs including Boeing and Airbus, holding Tier 1 supplier status with Boeing and Tier 2 status with Airbus. The manufacturing operations span CNC precision machining, sheet metal fabrication, surface treatment, and sub-assembly, making it a genuinely complex industrial environment from a data and process monitoring perspective.

2 Scope of Work

The primary focus of the internship was the design and analysis of data systems related to industrial emission monitoring, specifically in the context of the BEE PAT scheme reporting requirements applicable to the facility. The work involved examining how energy consumption data was currently being collected, aggregated, and reported across different production areas and equipment categories, identifying gaps between the data that was being collected and the analytical outputs that would be needed to support proactive compliance management, and exploring what additional data transformations and modelling approaches could bridge that gap.

Secondary work streams included exposure to the production planning systems used to schedule CNC machining operations, the quality management information systems that log inspection outcomes and non-conformance reports, and the supply chain data systems used to track raw material inventory and procurement costs. This broader exposure was valuable in understanding the ecosystem of data that an aerospace manufacturing facility generates and the integration challenges involved in bringing it together for analytical purposes.

3 Key Observations and Learnings

Several observations from the internship directly shaped the research design of this dissertation. First, the gap between data collection and data utilisation was larger than anticipated. The facility was generating continuous, high-quality energy consumption data through its building management and production monitoring systems, but this data was typically reviewed in monthly summary form by the environment, health and safety function rather than being used in real time by production planning. The opportunity to use the same data to predict compliance trajectories rather than just report historical actuals was clearly visible but had not been systematically pursued.

Second, the impact of production mix decisions on emission intensity was significant and not well understood quantitatively. Different machining operations consume very different amounts of energy: a heavy roughing cut on titanium in a five-axis machining centre consumes substantially more energy than a light finishing operation on aluminium. Similarly, surface treatment operations, particularly anodising and hard coat processes, have high energy intensities relative to their process time. When production planning teams made scheduling decisions, they were optimising for throughput, quality, and delivery performance without quantitative input about the emission intensity implications of different sequencing choices. A compliance risk tool that could translate production schedules into projected emission intensity trajectories, and flag sequences that would push the facility toward its regulatory threshold, would have genuine operational utility.

Third, the commodity price exposure at the procurement level was being managed without any systematic regime-awareness. The procurement function tracked historical price trends and used supplier relationship management to negotiate favourable terms, but it did not have access to any quantitative signal about whether current market conditions represented a price regime transition that warranted pre-emptive hedging or accelerated inventory positioning. The willingness of procurement managers to engage with the concept of regime-based price analytics, when it was discussed during the internship, suggested that the demand for such capability existed and that the barrier was access to the analytical framework rather than organisational receptiveness.

4 Connection to the Dissertation Research

The direct line from these internship observations to the research design of this dissertation is straightforward. The compliance risk module in the framework addresses the first and second observations: it transforms historical emission data into a real-time compliance risk signal, incorporates near-term production trajectory information through the rolling emission intensity trend feature, and would, in a full operational deployment, allow production planners to see the compliance implications of their scheduling decisions before they are made rather than after they have been executed.

The price regime prediction module addresses the third observation: it provides a quantitative, systematically updated signal about the current commodity price regime and the probability of a near-term transition, precisely the kind of forward-looking intelligence that the procurement function at a raw-material-intensive manufacturer needs but typically does not have. The ADWIN and Page-Hinkley drift detection layer addresses a concern that emerged from observing how models and systems in the facility became outdated without any systematic review process: as process configurations change, as regulatory thresholds are revised, and as market conditions evolve, any predictive model will degrade, and the facility needs a mechanism to detect that degradation and respond to it.

The Streamlit dashboard architecture was directly inspired by the observation that the most analytically sophisticated tool in the world is worthless if the people who need to use it cannot access and interpret its outputs in the flow of their operational day. The production planner, the procurement manager, and the environment, health and safety officer are not data scientists. A dashboard that presents regime probabilities, compliance risk levels, and drift alerts in a clear, colour-coded, threshold-based format — without requiring any programming knowledge to operate — is the appropriate interface layer for an operational risk tool in this context.

5 Reflection on the Value of Industrial Exposure

Looking back on the internship from the vantage point of having completed this dissertation, I think the most important thing it provided was a reality check on what research-driven tools need to do to be genuinely useful rather than technically impressive. Academic machine learning work tends to be evaluated on benchmark datasets against benchmark metrics, in an environment where the modelling assumptions are clean and the data is well-curated. Industrial deployment is different in almost every respect: the data is messier, the users are not statisticians, the consequences of a false alarm or a missed detection are real and sometimes costly, and the system needs to keep working reliably as conditions change without constant technical intervention.

The concept drift detection layer in this framework is, in a sense, the most direct response to that reality. It is the feature that acknowledges that the world will not stay the same as the model was trained on, and that a responsible system needs to monitor its own validity rather than assuming it. That insight came not from reading papers about concept drift but from watching what happened in practice when data distributions shifted in a manufacturing environment and nobody noticed until the predictions were already wrong. Building a system that notices for itself — and does something about it — is what this dissertation attempts to demonstrate is possible. Whether it succeeds is for the examiners and, eventually, for operational deployment to judge

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CHANDAN D

Bearing USN 22SSDS415012 *Has Completed The Internship*

Of Our Organization

This Certificate Is To Inform That The Student Has Completed

The Internship From 22-12-2025 *to* 06-02-2026 *in Bangalore*

06-02-026

Date

Kaladevaraj

Signature

motherson

Date: 5th May 2026

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that **Mr Chandan**, a student of **Bachelor of Science (Honours) in Data Sciences and Analytics** at **M. S. Ramaiah University of Applied Sciences, Bengaluru**, has successfully completed his internship **from 2nd March 2026 to 2nd May 2026.**

During the course of his internship, he worked on the project titled:

"An Integrated AI Framework for Industrial Risk Management: Predicting Raw Material Price Regimes and Carbon Compliance Risk Using Concept Drift Detection."

As part of this internship, he demonstrated strong technical and analytical capabilities in the domains of **machine learning, data science, and industrial risk analytics**. His work involved:

- Part of designing and implementing predictive models for **commodity price regime detection**
- Helped develop a **carbon compliance risk classification system** using machine learning techniques

Throughout the internship, Mr Chandan exhibited a high level of dedication, problem-solving ability, and a keen interest in applying artificial intelligence to real-world industrial challenges. His ability to integrate multiple AI components into a cohesive framework reflects strong research aptitude and practical implementation skills. He maintained professionalism, met project milestones, and actively contributed innovative ideas throughout the development process.

We wish him all the very best for his future.

From CIM TOOLS PRIVATE LIMITED,

Sathish K
Manager - Human Resources
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